How Turkic Is Anatolian Turkish?

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***Abstract*—How much of the modern Anatolian Turkish genome comes from the medieval Turkic populations of the steppe, and how much from the people who already lived in Anatolia when the Seljuks arrived? We model modern Turkish regional averages from the published Global-25 (G25) reference data as mixtures of Byzantine Anatolia (AD 500–1100), a medieval Turkic pool from Kazakhstan and Kyrgyzstan (AD 650–1200) and a Kartvelian source. The core of Anatolia (778 people from the western, central and northern provinces) fits as 54.9% Byzantine Anatolia, 23.1% medieval Turkic and 21.9% Kartvelian, and the core regions range from 20.6% to 28.2% Turkic. The author's own kit gives 23.0% on a commercial model. Replacing the steppe pool with the Çapalıbağ individuals of southwestern Türkiye (1300–1650 CE), the earliest Anatolian Turks with published DNA, improves every core fit. The core regions come out 56.9% to 77.2% Çapalıbağ-like, and an Ottoman-era source from Kaman-Kalehöyük gives the same Turkic share by a third route. An appendix fits all three models to 89 Turkish subgroups. Splitting the pre-Turkic side shows a Bronze Age Anatolian and Caucasian base with no Greek component. The closest modern populations to core Anatolia are Azerbaijanis, followed by western and Alevi Kurds and eastern Turks who share the Azerbaijani cluster; Greeks are no closer than Armenians, and the non-Oghuz Turkic peoples of Central Asia are distant.**

***Index Terms*—ancient DNA, admixture modeling, Global-25, Anatolia, Turkic peoples, Göktürk Khaganate, Seljuks, Çapalıbağ**

I. Introduction

Few questions start more arguments online in Türkiye than "how Turkic are the Turks?" Most answers are ideological. A narrower version of the question can be answered with genetics: what share of the modern Anatolian genome traces back to the medieval Turkic populations of the steppe, and what share to the population that was already in Anatolia when the Seljuks arrived after 1071?

We answer it with G25 principal-component coordinates and published ancient DNA, in four steps. Section II explains why the Turkic source starts at the Göktürk era. Section III describes the sources and the two control sites, and Section IV the fitting. Section V reports the fits for every Turkish region, what the pre-Turkic layer is made of, and where modern Anatolian Turks sit among other populations. The analysis scripts, the model definitions and the results are in the project repository.[1]

II. What "Turkic" Means in the Model

"Turkic" has no single genetic origin, and the record gets less certain the further back it goes. The first Turkic-speaking peoples attested in written sources are the Oghur-speaking tribes on the Pontic steppe in the 5th century CE. The first people to call themselves *Türk* are those of the Göktürk Khaganate, founded in 552. Before that, the names belong to confederations whose make-up is debated. The language or languages of the Xiongnu may have been Turkic, Mongolic, Yeniseian or a mixture, and the European Huns are just as unresolved. These confederations were political units, not genetic ones. There is therefore no clean proto-Turkic population to fit against, and we do not try.

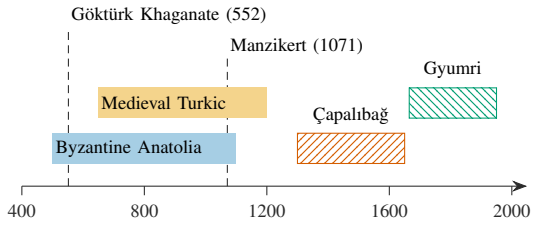


Fig. 1. Model sources and controls on a timeline. Shaded bands are the date windows of the two ancient sources; hatched bands are the two control sites from [3]. Vertical marks are the attested events used to anchor the Turkic source.

The Göktürks are a usable anchor. They are the first identifiable Turkic polity, and the medieval Turkic individuals of the Central Steppe and the Tian Shan represent their population genetically [1], [2]. These individuals were already mixed: they sit between ancient West Eurasian and East Eurasian populations. In this paper "Turkic" therefore means that Göktürk-era profile of AD 650–1200, not a pure steppe origin. After the Western Turkic Khaganate fell in the 7th century, the Oghuz moved west, and their descendants, the Seljuks, entered Anatolia after the battle of Manzikert in 1071 (Fig. 1).

III. Data

A. Reference data and targets

All population averages come from the published G25 reference data as used by Hanami DNA: 3,227 modern and 5,071 ancient rows. A group average weights each row by its sample size and leaves out outlier, low-resolution and contaminated rows. The targets are Turkish regional averages: West Anatolia (229 people), Central Anatolia (387), North Anatolia along the western Black Sea (175), the south around Adana, Mersin and Gaziantep (109), Trabzon, Rize and Artvin (48), Erzurum and the east (75) and the Balkan Turks (57). Core Anatolia is the first three together (778 people, including

[1] https://github.com/samliumay/how_turkic_is_anatolian_turkish

TABLE I
SOURCES OF THE MEDIEVAL MODELS

Source	Period	Rows
Medieval Turkic	AD 650–1200	Kazakhstan and Kyrgyzstan Early Medieval Turk, Karakhanid, Kipchak (8 people)
Byzantine Anatolia	AD 500–1100	Nicaea, Barcın, Ilıpınar, Samantas, Stratonikeia (26 people)
Early Anatolian Turkic	1300–1650	Çapalıbağ, SW Türkiye (12 people)
Ottoman Anatolian Turkic	Early Modern	Kaman-Kalehöyük, central Anatolia (3 people)
Kartvelian	modern	Georgian, Mingrelian, Svan, Laz

Istanbul's Anatolian-profile rows). For comparison we fit the Caucasus Azerbaijani average (134).

B. Sources

Table I lists the sources. The medieval Turkic pool holds the Early Medieval Turk, Karakhanid and Kipchak individuals from Kazakhstan and Kyrgyzstan [1], [4]. Byzantine Anatolia is the pre-Seljuk baseline. The Kartvelian source stands for the Caucasus-side component.

C. Controls

Two sites from the Southern Arc study [3] anchor the migration in time.

Çapalıbağ, southwestern Türkiye (1300–1650 CE). These individuals postdate the establishment of the Seljuk dynasty and carry substantial East Asian ancestry. The Y-chromosome lines include N-A9416, a Northeast Asian paternal line, and the mitochondrial lines include the East Eurasian A17 next to West Eurasian lines (W9, K1b1c, H13a1a1, HV14a, T2a1a, J2b2, H43, K1a). Their dates fall roughly 12 generations after the Seljuk expansion, and they sit halfway between Byzantine Anatolians and their Central Asian ancestors. They were descendants of real migrants who had already mixed with the local population. On our sources Çapalıbağ models as 47.3% Byzantine Anatolia, 36.7% medieval Turkic and 16.0% Kartvelian (fit 1.17), so we also use it as an early Anatolian Turkic source.

Kaman-Kalehöyük, central Anatolia (Early Modern). Three Ottoman-era individuals with Turkish and Turkic profiles. They model as 61.1% medieval Turkic, 27.3% Byzantine Anatolia and 11.6% Kartvelian (fit 1.74). With only three people, we use them as a second, later Anatolian Turkic source and read their results as a cross-check.

Gyumri, Black Fortress cemetery, Armenia (1665–1950 CE). One post-medieval individual, I18247, has a profile similar to present-day Azerbaijanis, unlike the Late Bronze Age individuals from the same site. It shows the Azerbaijani-like Oghuz profile in the Caucasus centuries ago.

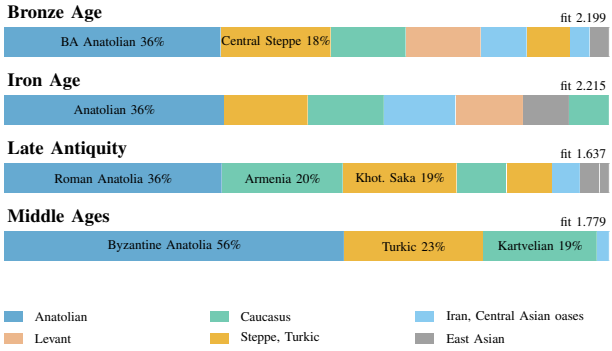


Fig. 2. The author's kit on Illustrative DNA, one model per period. Segments of 15% or more are labeled.

IV. METHOD

Each population is represented by its 25 scaled G25 coordinates. A target is modeled as a mixture of sources with non-negative weights that sum to one, chosen to minimize the Euclidean distance between the target and the mixture. The fit distance is that remaining distance times 100. Smaller is better, and it is only comparable between runs on the same coordinates. Distances between populations use the same scale. We fit three models to every target. Each combines Byzantine Anatolia and Kartvelian with one Turkic source: the medieval Turkic pool, Çapalıbağ, or Kaman-Kalehöyük. To split the pre-Turkic side, we fit the core average to Bronze Age Anatolia (EBA to MLBA), Mycenaean Greeks, La Tène Celts as a Galatian proxy, the Late Bronze Age Caucasus (Armenia) and the medieval Turkic pool, and repeat the fit over six source sets. The principal component analysis (PCA) is computed on 389 modern rows; ancient groups and the author are projected onto it.

V. RESULTS

A. The author's kit

Fig. 2 shows the author's kit on Illustrative DNA, one model per period. In the medieval model the kit is 56.2% Byzantine Anatolia, 23.0% Turkic, 18.8% Kartvelian and 2.0% Iranian Plateau (fit 1.779). In every earlier period the local Anatolian layer is about a third (35.8% to 36.4%), and the Caucasus and Iranian side is a quarter to a third.

B. Every region

Table II gives all three models for every region, and Fig. 3 draws the Çapalıbağ model. With the steppe pool, the core regions come out between 20.6% (Central) and 28.2% (West) Turkic, and the core average at 23.1% (fit 2.06). With Çapalıbağ as the source, the core regions are 56.9% to 77.2% Çapalıbağ-like, and every core fit improves (1.54 to 1.89 against 1.99 to 2.27). The two models agree: a core Anatolia that is 63.6% Çapalıbağ-like carries about 23% medieval Turkic ancestry through it, since Çapalıbağ itself is 36.7% medieval Turkic. The Ottoman-era source gives the same answer: core Anatolia is 38.0% Kaman-Kalehöyük-like (fit 1.86), and 38.0% of a source that is 61.1% medieval Turkic is again about 23%.

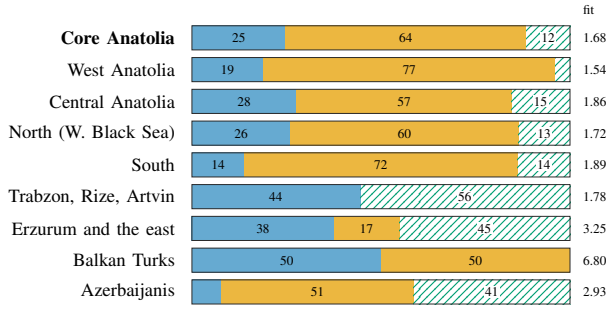


Fig. 3. Every region as Byzantine Anatolia (blue), Çapalıbağ as the early Anatolian Turkic source (orange) and Kartvelian (green, hatched), with the fit distance on the right.

Azerbaijanis get a similar Turkic share (18.2%) on a much larger Caucasus base (49.8%). For Trabzon and Erzurum the Caucasus source absorbs the signal, and the Balkan Turks fit badly in all three models. These three need their own source sets.

C. Subgroups

Table III in the appendix fits the three models to every Turkish subgroup row with at least three people, plus the Yörük, Çepni and Manav rows. The most Turkic rows are in the west and south: Muğla and Bolu reach 37% medieval Turkic, and the Yörük of Manisa, Mersin and Antalya 32% to 36%. Kayseri, Nevşehir, Kastamonu and Sivas sit lower, at about 17%. The northeast (Trabzon, Rize, Gümüşhane, Bayburt, Artvin), the Meskhetian Turks and the Turkish Cypriots come out below 3%. With these sources the model cannot separate their Caucasian or Levantine base from Turkic ancestry, so these low numbers do not measure it. Every Balkan row fits badly (fit above 4).

D. The pre-Turkic layer

Split into historical sources, the core average fits as 40.9% Bronze Age Anatolia, 33.7% Caucasus, 21.4% medieval Turkic, 4.0% Celtic and 0% Greek (fit 1.60). Across six source sets the Turkic share stays between 19.6% and 26.4%. The Greek source gets 0% in every set that includes Bronze Age Anatolia, and the Celtic share moves between 1% and 14%, too unstable to read. The pre-Turkic population was Anatolian, not Greek: it descends mostly from Bronze Age Anatolians and the Caucasus side. On the PCA (Fig. 5), Byzantine and Bronze Age Anatolia sit apart from mainland Greeks.

E. Where modern Anatolian Turks sit

Fig. 4 ranks populations by their distance from the core-Anatolian average. Azerbaijanis are closest (3.5 for the Caucasus average, 3.6 for Iran). The nearest Kurds are the western and Alevi Kurmanji of Malatya, Kayseri, Çorum and Kahramanmaraş (4.7), but they are closer still to Azerbaijanis (2.1) and to eastern Turks (2.2), so they belong to that eastern cluster. Kurds from the southeast, Iraq and Iran sit at 5.6. Anatolian and Pontic Greeks (7.0), North Caucasians and

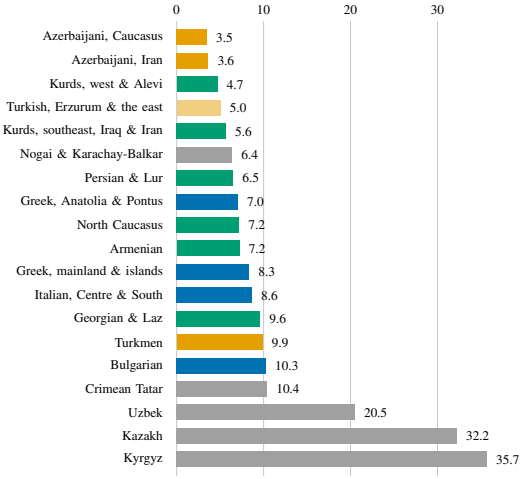


Fig. 4. G25 distance ($\times 100$) from the core-Anatolian Turkish average. Orange: Oghuz Turkic and Turkish; green: West Asia and the Caucasus; blue: Europe; grey: other Turkic peoples.

Armenians (7.2), mainland Greeks (8.3) and southern Italians (8.6) follow at similar distances. Turkmens, Oghuz but far more East Eurasian, sit at 9.9. Uzbeks are at 20.5, Kazakhs at 32.2 and Kyrgyz at 35.7.

The PCA shows the same picture (Fig. 5). Core Anatolian Turks sit just above the Azerbaijanis, North Caucasians and Kurds, and a step toward the steppe from them. Greeks, southern Italians and Bulgarians sit high on PC2. The Central Asian Turkic peoples stretch far out along PC1, and Çapalıbağ sits between core Anatolia and Turkmens. The author sits inside the core-Anatolian cluster.

VI. Limitations

G25 distance fitting compresses a genome into 25 dimensions and is a heuristic. The shares move when the source set changes, which is why we report several models and the range across regions. qpAdm on SNP data with explicit outgroups would be the stricter test. Byzantine Anatolia and the Kartvelian source are genetically adjacent, so the optimizer can trade one for the other. The medieval Turkic pool is itself part West Eurasian, so the strictly East Eurasian fraction in Anatolia is smaller than the Turkic share. The kit results come from a commercial service whose source panels are not published in full.

VII. Conclusion

The first anchor that can be identified genetically is the Göktürk-era steppe, itself a mixture of West and East Eurasian ancestry. The earlier confederations remain unresolved. The migration is visible in ancient DNA: Çapalıbağ carries East Asian ancestry about 12 generations after the Seljuk expansion. Core Anatolia is about a fifth to a quarter medieval Turkic, between 20.6% and 28.2% by region, and 56.9% to 77.2% Çapalıbağ-like. The pre-Turkic layer is Anatolian, not Greek, and descends mostly from Bronze Age Anatolians and the Caucasus side. The closest relatives of Anatolian Turks are the

TABLE II
THREE MODELS PER REGION: BYZANTINE ANATOLIA (BYZ), KARTVELIAN (KAR) AND ONE TURKIC SOURCE (%; FIT ×100)

	Medieval Turkic (650–1200)				Çapalıbağ (1300–1650)				Kaman-Kalehöyük (Ottoman)			
Target	Byz	Tur	Kar	Fit	Byz	Çap	Kar	Fit	Byz	Ott	Kar	Fit
Core Anatolia	54.9	23.1	21.9	2.06	24.8	63.6	11.6	1.68	44.5	38.0	17.5	1.86
West	55.4	28.2	16.4	1.99	18.9	77.2	3.9	1.54	42.8	46.2	11.0	1.76
Central	54.7	20.6	24.6	2.20	27.7	56.9	15.3	1.86	45.4	34.0	20.6	2.02
North (W. Black Sea)	54.7	22.0	23.2	2.07	26.0	60.5	13.4	1.72	44.8	36.2	19.0	1.88
South	48.1	26.4	25.5	2.27	13.8	72.4	13.8	1.89	36.3	43.3	20.4	2.06
Trabzon, Rize, Artvin	44.5	0.0	55.5	1.78	44.5	0.0	55.5	1.78	44.5	0.1	55.4	1.78
Erzurum and the east	45.8	5.8	48.4	3.34	37.6	17.2	45.3	3.25	43.0	10.0	47.0	3.28
Balkan Turks	81.1	18.9	0.0	6.88	50.1	49.9	0.0	6.80	69.4	30.6	0.0	6.84
Azerbaijanis (Caucasus)	32.0	18.2	49.8	3.24	7.8	50.9	41.3	2.93	23.7	30.1	46.2	3.13

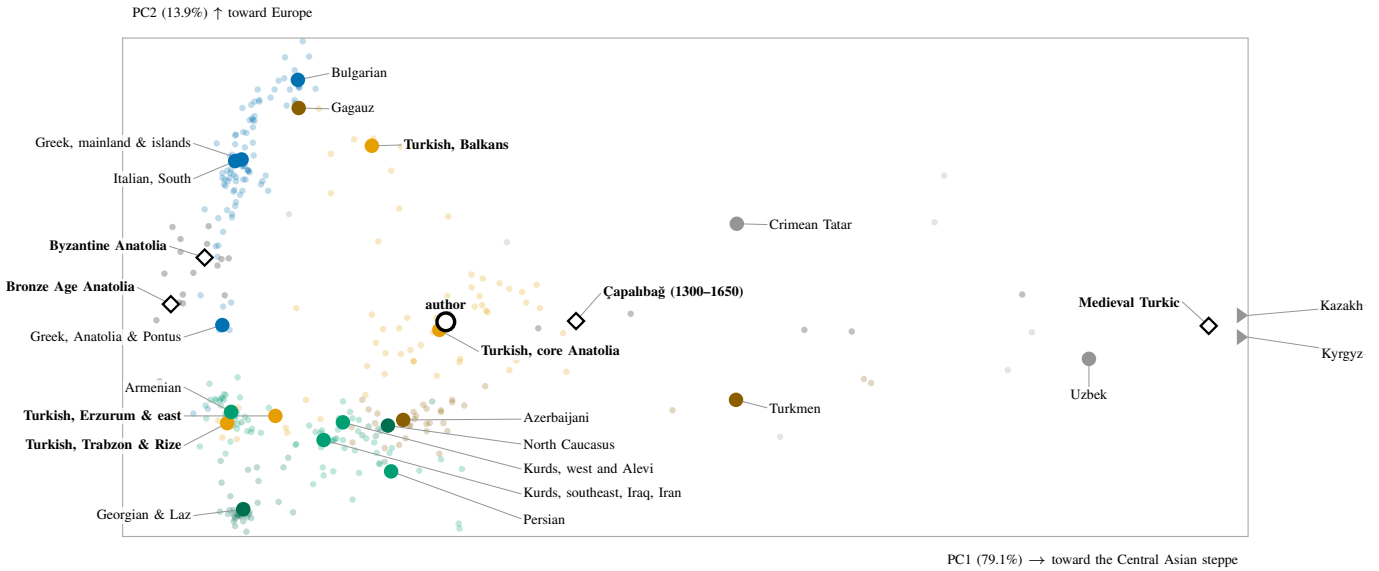


Fig. 5. PCA of modern West Asian, Caucasian, European and Turkic populations (PC1 79.1%, PC2 13.9% of the variance). Dots are group averages over faint individual rows; diamonds are projected ancient groups; the open circle is the author. Groups beyond the right edge are marked with an arrow.

Azerbaijanis; Greeks are no closer than Armenians, and the non-Oghuz Turkic peoples of Central Asia are far away.

DATA AVAILABILITY

The analysis scripts, the model definitions and their results are in the project repository (footnote on the first page). The G25 reference data are published, and the ancient individuals are published in [1]–[4]. The author's own coordinates are not shared.

ACKNOWLEDGMENT

An AI assistant (Claude, Anthropic) helped draft and edit the text of this paper and write the analysis scripts. The author chose the models and the sources and is responsible for the content.

APPENDIX A
ALL TURKISH SUBGROUPS

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TABLE III

TURKIC SHARE (%) OF EVERY TURKISH SUBGROUP ROW ON THE THREE MODELS: MEDIEVAL TURKIC (TUR), ÇAPALIBAĞ (ÇAP) AND KAMAN-KALEHÖYÜK (OTT), EACH WITH BYZANTINE ANATOLIA AND KARTVELIAN. FIT IS THE BEST OF THE THREE. ROWS FROM THE HANAMI DNA REFERENCE DATA; n IS THE NUMBER OF PEOPLE.

Subgroup	n	Tur	Çap	Ott	Fit	Subgroup	n	Tur	Çap	Ott	Fit
Adana	28	28	76	45	1.74	Istanbul (Anat.)	13	21	58	34	1.50
Afyonkarahisar	18	25	68	40	2.08	Istanbul (mixed)	4	22	60	36	5.60
Afyonkarahisar Northeast	28	26	70	42	2.02	Izmir	9	28	75	45	1.41
Aksaray	23	21	58	35	2.07	Izmir Cepni	2	30	82	49	2.47
Amasya	12	18	50	30	2.33	Kahramanmaras	25	21	58	35	2.28
Ankara	27	26	72	43	1.83	Karabuk	8	24	67	40	1.94
Ankara Manav	1	32	86	52	2.31	Karaman	20	19	54	32	1.84
Ankara Sereflikochisar	11	22	62	37	2.62	Kars	4	11	30	18	3.66
Antalya	12	33	90	54	1.02	Kastamonu	24	17	46	28	2.10
Antalya Yoruk	2	32	88	53	1.78	Kayseri	35	17	47	28	1.46
Ardahan	3	15	43	25	2.90	Kirikkale	6	25	68	41	2.03
Artvin	7	2	6	3	3.73	Kirsehir	20	25	68	41	2.19
Aydin	31	28	78	47	1.33	Kocaeli	5	27	75	45	2.69
Balikesir	17	31	84	50	1.75	Konya	14	22	61	36	1.11
Balikesir Cepni	2	35	93	57	2.84	Kutahya	17	28	77	46	1.93
Balkans (mixed)	11	21	54	33	5.96	Malatya	19	12	34	20	3.12
Bartin	8	23	63	38	1.70	Manisa	9	27	74	44	2.29
Bayburt	14	2	7	4	3.13	Manisa Cepni	1	30	83	50	2.29
Bilecik	4	28	75	45	2.25	Manisa Yoruk	9	35	96	58	1.52
Bolu	14	37	100	61	1.80	Mersin	19	34	92	55	1.66
Burdur	16	28	77	46	1.66	Mersin Yoruk	3	36	98	59	1.49
Bursa	6	28	77	46	2.37	Meskhetian Ardahan	5	2	6	4	1.16
Canakkale Biga	3	30	81	49	2.01	Meskhetian Georgia	4	2	5	3	1.88
Cankiri	12	20	56	33	2.09	Mugla	12	37	100	61	1.46
Central Macedonia	3	15	38	24	6.20	Nevsehir	32	17	46	28	2.11
Corum	19	19	53	32	2.49	Nigde	10	18	48	29	2.25
Cyprus	33	3	8	5	2.05	North Macedonia	15	14	37	22	8.13
Denizli	9	31	85	51	1.15	North Thrace	52	23	62	38	6.42
Duzce	4	32	88	52	1.73	Northern Bulgaria	17	23	60	37	7.05
East Thrace Gelibolu	3	21	56	34	4.27	Ordu	30	23	63	38	1.89
Eastern Macedonia	3	17	44	27	6.66	Rize	5	0	0	0	1.98
Elazig	4	11	31	18	3.01	Samsun	16	20	55	33	1.67
Erzincan	16	9	26	15	3.22	Sinop	11	19	53	31	1.78
Erzurum	12	6	18	11	2.35	Sinop Northwest	10	9	25	15	2.12
Erzurum Central	11	7	20	12	3.95	Sivas	63	17	48	28	2.42
Erzurum North	15	2	6	3	3.93	Thessaly	3	16	41	25	6.34
Eskisehir	9	28	76	46	1.98	Tokat	21	20	55	33	2.03
Gaziantep	19	22	60	36	2.45	Trabzon	34	0	0	0	1.77
Giresun	18	32	86	52	1.10	Tunceli	3	9	26	15	3.65
Giresun North	44	31	86	51	2.16	Usak	21	27	74	44	1.83
Giresun South	10	15	42	25	2.72	West Thrace	3	28	73	45	5.12
Gumushane	3	0	1	1	1.61	Western Macedonia	5	12	32	19	7.69
Hatay	13	27	75	45	2.12	Yozgat	30	20	56	33	2.07
Igdir	5	17	49	29	3.66	Zonguldak	17	28	77	46	1.87
Isparta	34	19	53	32	2.08						